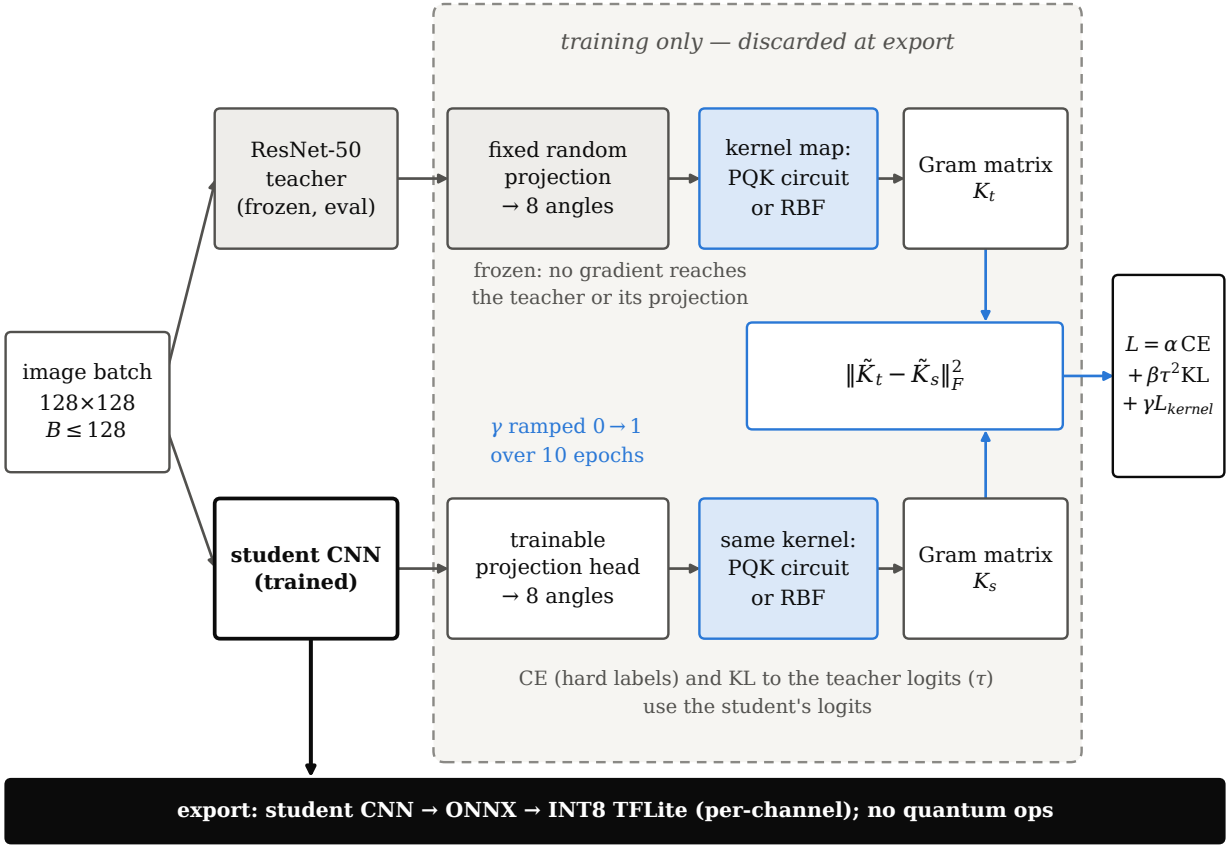
training only — discarded at export
image batch
128×128
$B \le 128$
ResNet-50
teacher
(frozen, eval)
fixed random
projection
→ 8 angles
kernel map:
PQK circuit
or RBF
Gram matrix
K_t
frozen: no gradient reaches
the teacher or its projection
$\|\tilde{K}_t - \tilde{K}_s\|_F^2$
$L = \alpha\,\mathrm{CE}$
$+ \beta\tau^2\mathrm{KL}$
$+ \gamma L_{kernel}$
γ ramped 0 → 1
over 10 epochs
student CNN
(trained)
trainable
projection head
→ 8 angles
same kernel:
PQK circuit
or RBF
Gram matrix
K_s
CE (hard labels) and KL to the teacher logits (τ)
use the student's logits
export: student CNN → ONNX → INT8 TFLite (per-channel); no quantum ops